AI-POWERED SMART ORCHID CARE SYSTEM FOR VANDA CULTIVATION

COMMON INTEGRATED SOLUTION REPORT

Team member	Student ID	Individual component
Wickramasinghe D. P.	IT22065230	Component 1 - Hybrid Pollination and Compatibility Analysis
Rathnamalala D. M. J. P.	IT22064936	Component 2 - Disease Detection and Severity Grading
Vithanage P. V. P. M.	IT22062642	Component 3 - Smart Watering and Automated Fertilisation
Samadinee L. D. N.	IT22083500	Component 4 - Growth Stage Classification and Bloom Timing

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Honours) in Information Technology

Specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. We have disclosed all use of generative AI and other automated tools in accordance with the module policy. We remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this report.

Student name	Student ID	Signature	Date
Wickramasinghe D. P.	IT22065230		10/10/2026
Rathnamalala D. M. J. P.	IT22064936		10/10/2026
Vithanage P. V. P. M.	IT22062642		10/10/2026
Samadinee L. D. N.	IT22083500		10/10/2026

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	Signature	Date
Supervisor: Ms. Hansi De Silva		
Co-supervisor: Ms. Thilini Jayalath		

ABSTRACT

Vanda orchids are one of Sri Lanka's higher-value floricultural exports, yet almost every decision that keeps a plant productive is still taken by unaided visual judgement in the shade house. Four of those decisions carry a measurable cost when they go wrong: an undiagnosed leaf lesion spreads through a house in days, water delivered at the wrong hour scorches exposed roots, a misread growth stage wastes a fertiliser cycle, and pollinating an unfit plant wastes the nine months a Vanda seed pod needs to mature. Published smart-agriculture systems address these tasks separately and commonly report a single accuracy figure with no baseline, so a model that merely predicts the commonest class can appear competent. This project integrates four machine-learning components into one mobile system sharing a single FastAPI backend and Firebase database, and reports every figure beside the majority-class baseline of the same held-out split. Disease classification reaches 0.806 accuracy and 0.778 macro F1 on 67 held-out leaves against a 0.418 baseline. Growth stage reaches 0.940 across six classes against 0.196, and bloom timing is predicted to within 0.92 days. Watering time is predicted to within 8.5 minutes with an out-of-time coefficient of determination of 0.951. Pollination suitability reaches 0.717 against 0.639, validated by StratifiedGroupKFold grouped by plant after an image-level split was found to leak. A two-stage input gate refuses photographs that are not orchids before any verdict is issued. Weaker results are reported rather than hidden: severity grading reaches only 0.450 macro F1. The contribution is an integrated system whose every claim is stated against a baseline and whose failure modes are documented.

Keywords: precision agriculture, orchid cultivation, transfer learning, out-of-distribution detection, integrated system evaluation

ACKNOWLEDGEMENTS

We thank our supervisor, Ms. Hansi De Silva, and co-supervisor, Ms. Thilini Jayalath, for their guidance throughout this project, and for pressing us repeatedly on the difference between a result and a measured result.

We are grateful to the orchid growers who gave us access to their shade houses and allowed us to photograph their plants over several months. We owe a particular debt to the breeder at Peradeniya whose account of intergeneric crossing contradicted the published compilations we had been reading, and who thereby prevented us from reporting a success rate that the available data could never have supported.

TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT 1

1.1 Background and context 1

1.2 Stakeholders and user needs 2

1.3 Alignment with the Sustainable Development Goals 3

1.4 Purpose and scope of the common report 3

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 5

2.1 Overall problem definition 5

2.2 Root causes and overall research gap 5

2.3 Subproblem and component mapping 6

2.4 Overall aim and group objectives 7

2.5 Scope assumptions and constraints 8

3 OVERALL SOLUTION STRUCTURE 9

3.1 Integrated solution overview 9

3.2 High level architecture 9

3.3 Component 1 summary 10

3.4 Component 2 summary 12

3.5 Component 3 summary 14

3.6 Component 4 summary 17

3.7 End to end user and data flow 20

4 COMPONENT INTEGRATION AND COHESIVE OPERATION 21

4.1 Integration strategy 21

4.2 Interfaces data contracts and shared models 21

4.3 Shared nonfunctional requirements 23

4.4 Integrated workflow and exception handling 24

4.5 Team contribution and dependency matrix 25
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 26
5.1 Common technology stack and infrastructure 26
5.2 Development and integration process 27
5.3 Deployment and operational configuration 27
5.4 Evidence of team execution 28
6 INTEGRATED TESTING EVALUATION AND VALIDATION 30
6.1 Evaluation framework and success criteria 30
6.2 Integration and end to end testing 30
6.3 Overall results 31
6.4 Stakeholder and user validation 36
6.5 Achievement of common objectives 37
6.6 Integrated limitations and threats 38
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 39
7.1 Value proposition and beneficiaries 39
7.2 Market alternatives and differentiation 39
7.3 Business model and feasibility 40
7.4 Adoption intellectual property and compliance 41
7.5 Sustainable Development Goal contribution 42
8 CONCLUSION AND FUTURE DIRECTIONS 43
8.1 Overall conclusion 43
8.2 Principal integrated contribution 43
8.3 Future directions 44
REFERENCES 46
APPENDICES 48

Appendix A Integrated interface specification 48

Appendix B Team contribution declaration 49

LIST OF FIGURES

Figure 3.1: Deployed architecture of the integrated solution. Image evidence and sensor evidence enter separately; four components share one backend, one database and one application, and each returns a single verdict to the grower 10
Figure 3.2: Component 1's three-stage assessment path. The two gates exist because a classifier trained only on orchids will extrapolate confidently on any image presented to it; refusal is an answer the system is designed to give 11
Figure 6.1: Classification accuracy of three components against the majority-class baseline of the same held-out split. The distance between the bar and the marker is the evidence of learning 33
Figure 6.2: Component 1 model selection. All four candidates scored under StratifiedGroupKFold grouped by plant; the support vector machine and the random forest fall below the majority-class baseline 34
Figure 6.3: Out-of-fold confusion matrix for Component 1's deployed classifier, with counts and row percentages 35

LIST OF TABLES

Table 2.1: Mapping of the four subproblems to components, with the run-time dependencies that make the solution integrated rather than parallel..... 6

Table 4.1: Interfaces that cross a component boundary, with the error behaviour each consumer may rely on 22

Table 4.2: Division of research, development and integration work across the four component owners. Entries marked for completion are to be filled by the owner concerned 25

Table 6.1: Headline results for the integrated system, each stated beside the baseline or naive error of the same split..... 32

Table 6.2: Component 1 input gate measured on a held-out validation set of 455 non-orchid and 88 unseen orchid photographs. Grouped cross-validation on the project's own photographs gives a false-refusal rate of 2.2% 35

Table 6.3: Evidence for each group objective and the outcome claimed..... 37

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
CNN	Convolutional Neural Network
ERA5	Fifth-generation European Centre for Medium-Range Weather Forecasts Reanalysis
ExG	Excess Green index
F1	Harmonic mean of precision and recall
LKR	Sri Lankan rupee
MAC	Media Access Control
MAE	Mean Absolute Error
REST	Representational State Transfer
RHS	Royal Horticultural Society
RTDB	Realtime Database
SDG	Sustainable Development Goal
TLS	Transport Layer Security
VPD	Vapour Pressure Deficit

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Sri Lanka's floriculture sector is small by export volume but unusually dependent on skill. Vanda orchids sit at the upper end of it: a mature flowering plant commands a price that a vegetable crop of the same bench area could not approach, and the plants are grown overwhelmingly by smallholders in shade houses of a few hundred square metres rather than on industrial nurseries. The economics of that arrangement are unforgiving. A grower with two hundred plants has no margin for a disease that moves through a bench unnoticed, and no second chance at a breeding season that has already been spent.

What makes the work difficult is not that the right decisions are unknown. Vanda husbandry is well documented, and an experienced grower reads a plant accurately in a few seconds. The difficulty is that the knowledge lives in people rather than instruments, it is acquired over years, and it is applied to every plant individually and from memory. A grower who is tired, new, or simply managing more plants than can be inspected daily will miss things, and the feedback that would correct the mistake often arrives months later.

Four decisions dominate. Whether a leaf marking is a spreading infection or cosmetic damage determines whether a plant is isolated today or in a week. Which phenological stage a plant has reached determines the fertiliser formulation it should receive, and feeding a dormant plant is actively harmful. When and for how long to water determines whether exposed aerial roots are hydrated or scorched, and the correct answer moves with the weather rather than the clock. Finally, whether a plant is fit to carry a seed pod determines whether nine months of a breeding programme produce a capsule or nothing at all. These four decisions are the subject of this project.

1.2 Stakeholders and user needs

The primary stakeholder is the smallholder Vanda grower who manages the shade house personally. This user is not a technician. The relevant design consequence is that the system must produce an action rather than a number: "isolate this plant and treat it first" is usable, whereas a vector of class probabilities is not. It must also work on the hardware the grower already owns, which in practice means an Android phone and an intermittent connection.

Secondary stakeholders include commercial nurseries that run breeding programmes and would benefit most from the pollination component, agricultural extension officers who advise several growers and need a record they can review remotely, and exporters whose quality claims depend on consistent plant condition. A fourth group, which the team treats as a stakeholder rather than a user, is the breeding community whose accumulated crossing experience is currently undocumented and therefore unavailable to anyone who did not personally acquire it.

Engagement with growers shaped the system more than once. The most consequential instance concerned watering: the original design asked whether a plant needed water, inferred from a root-moisture probe. A grower pointed out that probing the roots of every plant is impractical and that daily watering is mandatory in any case, which meant the model was answering a question nobody asks. The question was reformulated as what time today and for how long, and the component was rebuilt around it. A second instance is recorded in Section 2.2.

For Component 3, the consultation was a site visit to a commercial orchid farm on 17 June 2026 and informal conversations with the grower who owns it. The grower described watering the whole farm every day as hard work, and said that control of the whole farm at once, watering several sections in one run and a history of waterings would all be useful, and the component now offers each of these. No formal study was run and no interview instrument was used. The same grower later gave access for the sensor nodes described in Section 6.4.

1.3 Alignment with the Sustainable Development Goals

The project contributes most directly to Sustainable Development Goal 12, responsible consumption and production, through target 12.2 on the efficient use of natural resources. Component 3 replaces one watering routine for a whole house with a schedule for each section, set by that section's measured air conditions, and it feeds only with water, at reduced strength and never to feed dormant plants at all. Both reduce the quantity of water and nutrient applied for the same horticultural outcome, and the watering schedule also bears on Goal 6, clean water and sanitation, through target 6.4 on water use efficiency. The claim the team is willing to defend is that the system changes how much is applied and when it has not yet measured the resulting saving in a working shade house over a season, and does not claim one.

A secondary contribution is to Goal 8, decent work and economic growth, through target 8.2 on productivity through technological upgrading. The relevant mechanism is narrow but real: the system transfers judgements that currently require years of experience into a form a less experienced grower can act on, which lowers the barrier to entering a high-value crop.

The team deliberately does not claim a contribution to Goal 2. Vanda orchids are an ornamental export crop, not food, and presenting horticultural automation as food security would be an overstatement of the kind this report is otherwise at pains to avoid.

1.4 Purpose and scope of the common report

This report presents the integrated problem and the integrated solution. It describes how the four components were partitioned, how they were joined into one system, how that system was evaluated as a whole, and what the complete solution is and is not able to do. It is written to be read on its own.

It is not a collection of four theses. The detailed derivations, the full experimental records, the ablations and the extended literature treatment for each component belong to the four individual theses submitted alongside it. Where a component-level

result is quoted here it is quoted because the integrated argument depends on it, and the reader is directed to the relevant thesis for the method that produced it.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is stated as follows. In Sri Lankan Vanda cultivation, the four recurring decisions that determine plant health and breeding outcome (disease diagnosis, growth-stage judgement, irrigation timing and parent selection) are made by unaided visual inspection, which is inconsistent between growers, unrecorded, and in several cases cannot be corrected before the loss has already occurred.

Two features of that statement matter. The first is that it is a problem of judgement under time pressure, not a problem of ignorance a solution that merely presents horticultural information does not address it. The second is that the four decisions interact. A plant misread as healthy is watered and fed on the schedule for a healthy plant and may be selected as a breeding parent, so an error in one judgement propagates into the others. That interaction is the reason the solution is one system rather than four tools.

2.2 Root causes and overall research gap

Three causes underlie the problem. The shade house carries no instrumentation, so no condition is recorded and nothing can be reviewed after the fact. Expertise is distributed thinly and transfers slowly, by apprenticeship rather than documentation. And the feedback loop is long: the consequence of a wrong pollination decision appears after roughly nine months, by which time the conditions that caused it cannot be reconstructed.

The research gap is narrower and more specific than the general observation that agriculture could be automated. Reviewing comparable published systems, the team found three recurring weaknesses. Tasks are addressed in isolation, so the interaction described above is never modelled. Accuracy is reported without a majority-class baseline, which makes a result uninterpretable whenever the classes are unbalanced.

And validation is frequently performed in a way that does not reflect deployment, most often by splitting at the level of the image rather than the plant or the day.

The team encountered the third weakness in its own work before recognising it in the literature. An early version of the pollination classifier reported perfect accuracy on a random image-level split. Because the 357 photographs came from only 28 physical plants, the split placed several images of the same plant on both sides, and the classifier was recognising individual plants rather than their condition. Regrouping the validation by plant reduced the same model to 0.717. That experience is the origin of the evaluation discipline applied across all four components and described in Section 6.1.

A second, independent gap concerns evidence rather than method. The authoritative record of orchid hybrids registers crosses that were made and succeeded. Nobody registers an attempt that produced no pod. The register therefore has no denominator, and the contradiction this creates is visible in the source material: a compilation derived from the register describes intergeneric crossing as moderately to highly successful, while a breeder at Peradeniya reports that most intergeneric crosses do not work. Both accounts are honest. The register sees the survivors the breeder sees the attempts. Any success percentage derived from the register alone would be survivorship bias presented as a measurement, and Component 1 is designed around that constraint.

2.3 Subproblem and component mapping

The common problem decomposes into four subproblems that do not overlap in scope but do depend on one another at run time. Table 2.1 states each subproblem, the component that addresses it, and the dependency that makes it part of a system rather than a standalone tool.

Table 2.1: Mapping of the four subproblems to components, with the run-time dependencies that make the solution integrated rather than parallel

Subproblem	Component	Primary input	Dependency on other components
Is this plant fit to breed from, and with which partner?	C1 Hybrid pollination and compatibility	Plant photograph, parent names	Consumes disease severity from C2 and flowering state from C4

Subproblem	Component	Primary input	Dependency on other components
Is this plant diseased, and how badly?	C2 Disease detection and severity	Leaf photograph	Supplies a severity signal consumed by C1; informs isolation before C3 waters a bench
When and for how long should this section be watered and when should it be fed?	C3 Smart watering and fertilisation	Sensor stream, weather forecast	None at run time. The growth stage that selects the feed is set by the grower. Shares the application, backend server, database and account system with the other three
What stage has this plant reached, and when will it bloom?	C4 Growth stage and bloom timing	Plant photograph, ambient readings	Determines the fertiliser formulation C3 applies; bloom timing bounds the window C1 can act in

The decomposition is deliberately organised by decision rather than by technology. Three of the four components use image classification, and it would have been possible to divide the work by technique instead. That would have produced components with no natural interface to one another, because a grower does not experience the shade house as a set of model families.

2.4 Overall aim and group objectives

The aim of the project is to deliver a single mobile system that converts the four recurring Vanda husbandry decisions from unaided visual judgement into recorded, measured recommendations, and to report the reliability of each recommendation against an explicit baseline.

Five group objectives follow from that aim. G1: deliver four predictive components covering disease, growth stage, irrigation and pollination, each trained and evaluated on data held out from its own training set. G2: join the four through one backend, one database and one application such that a verdict produced by one component is available to the others. G3: report every quantitative claim beside the majority-class baseline or held-out error of the same split, and publish results that fall short of it. G4: ensure the system refuses inputs it cannot interpret rather than producing a confident answer to an invalid question. G5: establish an evaluation protocol whose splits reflect the deployment condition, grouping by plant or by time where an image-level split would leak. Section 6.5 states the evidence for each.

2.5 Scope assumptions and constraints

The solution is scoped to Vanda and to lowland Sri Lankan growing conditions. Component 3's stress calibration spans five climate zones; Nuwara Eliya was downloaded and then deliberately excluded, because Vanda are not grown at its temperatures and its inclusion dragged the percentile scale so low that an ordinary lowland day was scored as extreme. The models themselves take no location feature, so the system transfers to any lowland site once coordinates are configured, but that claim has not been verified outside the zones trained on.

Three assumptions are load-bearing. The grower is assumed to be able to photograph a plant in daylight at roughly arm's length; images taken under coloured artificial light are outside the trained distribution and the input gate will usually refuse them. The sensor node is assumed to report honestly or not at all, with a failed channel reported as -999 and then dropped, never replaced by a default so that no decision is made from it. And the outdoor-to-shade-house weather conversion is a modelled transformation, not a measured one; validating it against the physical sensors is outstanding work and is stated as such.

The principal exclusions are the laboratory stages of breeding, which the system does not touch, and any claim about the viability of seed produced by a cross it has recommended. Component 1 ends at the decision to attempt a pollination. What happens in the flask afterwards is outside both the data and the scope.

3 OVERALL SOLUTION STRUCTURE

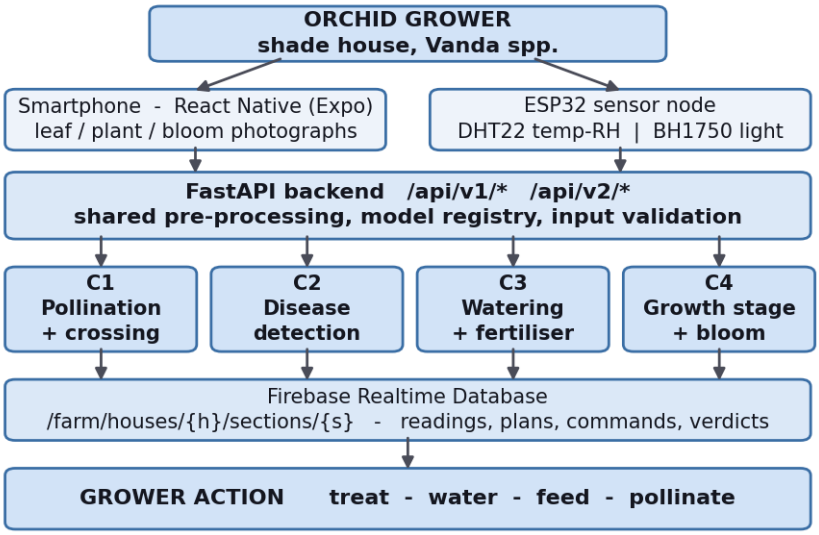
3.1 Integrated solution overview

The solution is a single mobile application backed by one server. The grower photographs a plant, or a sensor node reports a reading, and the system returns the one decision that evidence supports: treat, water, feed, or pollinate. Four models stand behind those four verdicts, but the grower encounters one application with four tabs and a shared record of what each plant has been told.

The value proposition is not that any individual model is novel. It is that the four judgements are produced from the same evidence base, recorded against the same plant, and reported with the same evidential discipline, so that a grower can see why a recommendation was made and an examiner can see how far to trust it. A system that answered three questions well and the fourth confidently but wrongly would be worse than useless in a shade house, which is why the refusal behaviour described in Section 4.4 is treated as a feature of the integrated solution rather than an implementation detail.

3.2 High level architecture

Figure 3.1 shows the deployed architecture. Two kinds of evidence enter independently: photographs from the grower's phone and ambient readings from ESP32 sensor nodes. Both arrive at a single FastAPI backend that owns pre-processing, the model registry and input validation. The four components are routers within that backend rather than separate services, which keeps the shared pre-processing honest: every component segments an image the same way because they call the same function.



Automation engine (60 s tick) closes the loop for C3; the other three return a verdict to the app.

Figure 3.1: Deployed architecture of the integrated solution. Image evidence and sensor evidence enter separately; four components share one backend, one database and one application, and each returns a single verdict to the grower

State lives in a Firebase Realtime Database in which each farm is a separate tenant, organised as farm, house and section, where a section is one microclimate zone. A section's latest reading, today's plan and its tray, fertiliser and control state are written under its section key, its reading history sits in a branch of its own, and pump commands go to the queue of the house's master controller. A component that needs another component's output reads it from the database rather than calling across. Component 3 additionally runs a background automation loop on a sixty-second tick, which is what turns a watering plan into a pump command, and a pour counts as done only when the master controller acknowledges it. The other three components are request-driven and return their verdict to the application.

3.3 Component 1 summary

Component 1 is owned by Wickramasinghe D. P. (IT22065230). It addresses the first subproblem: deciding whether a given plant is fit to breed from, whether a proposed pairing of two named parents has any precedent behind it, and what actually happened when the cross was attempted. Its inputs are a whole-plant photograph of each parent and, for the second question, the two parent names already written on nursery tags. Its outputs are a three-class suitability verdict with a confidence, an

evidence tier for a proposed cross, and a dated record of the attempt that the grower can return to as the capsule develops.

The core method operates in three stages, shown in Figure 3.2. A frozen ResNet18 pretrained on ImageNet converts the photograph into a 512-dimensional embedding; the 1000-way classification head is discarded, because it is of no use here and labels a Vanda flower a sea anemone with high confidence. The first stage is a one-class novelty detector over that embedding, which refuses an image whose cosine distance to its tenth nearest neighbour among 1,190 reference photographs exceeds 0.25. The second separates orchid blooms from other flowers. Only an image that passes both reaches the third stage, where 66 image and trait features are classified by XGBoost into Suitable, Moderate or Not Suitable.

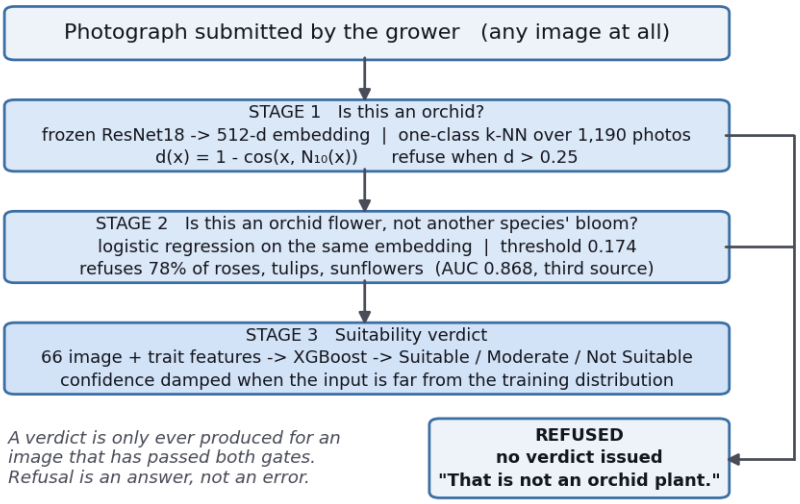


Figure 3.2: Component 1's three-stage assessment path. The two gates exist because a classifier trained only on orchids will extrapolate confidently on any image presented to it; refusal is an answer the system is designed to give

The second question is answered differently and deliberately so. Crossing compatibility is resolved from the registered hybrid record rather than from images, because the thing being asked about, whether two lineages have ever produced viable offspring, is not visible in a photograph. Resolving the name is the first step, and the component reports which of three routes it took: the name may state its genus, it may state none but appear in the knowledge base, or it may do neither, in which case the collection's genus is assumed and the result says so. A hybrid genus name is expanded into the natural genera it contains before any comparison is made, since

such a name is itself the record of a cross between those genera. The component returns an evidence tier and the precedents behind it, never a success percentage, for the reason given in Section 2.2. Direction is preserved throughout: by breeding convention the pod parent is named first, and A crossed with B is a different attempt from B crossed with A.

Component 1's contribution to the overall solution is threefold. It supplies the breeding decision that none of the other three address. It supplies the input-validation discipline that the whole system now relies on: the two-stage gate was built for Component 1 after the classifier returned a confident suitability verdict for a photograph of a laptop, and it is the mechanism by which the integrated system declines to answer questions it has no basis for answering. And it supplies the crossing logbook, which is the only part of the integrated system that collects outcomes rather than producing predictions. The registered record holds successes alone, so the failures a success rate would require have to be gathered.

3.4 Component 2 summary

Component 2 is owned by Rathnamalala D. M. J. P. (IT22064936). It addresses the second subproblem: whether a plant is diseased, with which disease, and how badly. Its input is a single leaf photograph, taken with the grower's phone camera or chosen from the gallery. Its outputs are a disease verdict with a confidence, a severity grade of mild, moderate or severe when a disease is identified, and a treatment recommendation keyed to the disease and the grade.

The core method is a cascade in which each stage can end the request (Figure 3.3). The first stage measures how unlike the training photographs an image is, as its mean cosine distance to the five nearest training images in the 1,280-dimensional feature space of the trained network [14], and refuses it beyond a distance of 0.55. The gate exists because a softmax classifier must always choose one of its trained classes: before it was added, a photograph of a meal was classified as healthy with 99.9% confidence. The second stage is a MobileNetV2 classifier [15] pretrained on ImageNet [16]. It was trained first as a new head on the frozen network and then by fine-tuning its top layers, with class weights to offset the 1.8-to-1 imbalance between

the two diseases. The third stage reports any prediction below 0.70 confidence, a threshold chosen on the validation split, as an unidentified condition and advises expert review. Only an identified disease reaches the fourth stage, a separate MobileNetV2 classifier that grades severity by the proportion of leaf area affected: under 10%, 10 to 40%, and over 40%. The grade then selects the treatment from a rule-based knowledge base.

On 67 held-out test leaves the classifier reaches 0.806 accuracy and 0.778 macro F1, against a majority-class baseline of 0.418. All 24 healthy test leaves were recognized as healthy, and ten of the thirteen errors are confusions between the two leaf-spot diseases, the distinction a non-specialist also finds hardest. MobileNetV2 was selected under identical conditions on the validation split, never the test split. EfficientNetB0 [1] scored 0.832 macro F1 against MobileNetV2's 0.822, a difference of a single validation image in accuracy, at 1.8 times the model size, while a network trained from scratch reached only 0.603, which is the evidence that transfer learning was necessary with 533 training photographs. Severity grading is weaker. On 43 held-out diseased leaves it reaches 0.465 accuracy and 0.450 macro F1 against a majority-class baseline of 0.395, although 90.7% of grades fall within one step of the true grade. The shortfall has identified causes: the photographs were split by disease before severity was labelled, leaving severe leaves at 45% of the training split but 21% of the validation split, and the severity network was trained with its pretrained layers frozen.

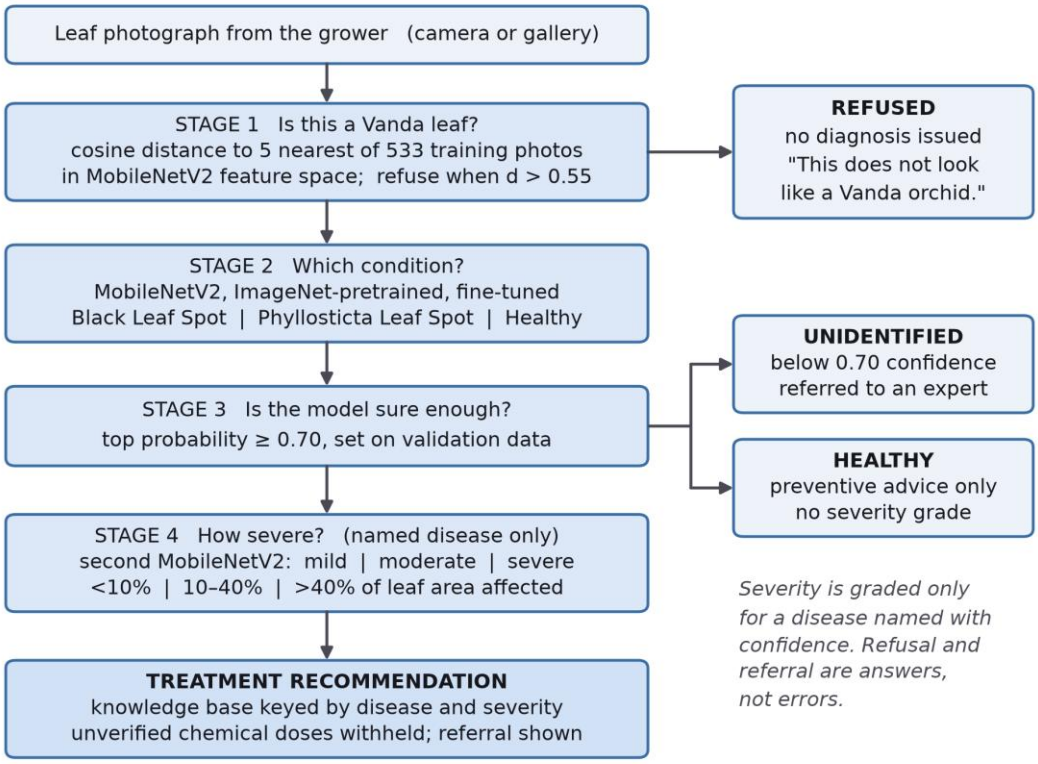


Figure 3.3: Component 2's four-stage diagnosis path. Each stage can end the request; severity is graded only for a disease identified with confidence, and refusal or referral is returned as an answer rather than an error

Component 2's contribution to the overall solution is the disease signal that Component 1 consumes at run time through the provider interface in Table 4.1. The component also applies the system's refusal principle to diagnosis at two levels, refusing an unfamiliar image before classification and referring a low-confidence classification to an expert. It withholds what it cannot yet support: no chemical dose is displayed until it has been verified against the product label and the Sri Lankan Registrar of Pesticides' registered list, and a referral message is returned in its place. Photographs of conditions the system does not recognize can be submitted from the application. They are stored as originals in a review queue, never change the deployed model directly, and make a new class eligible for training only at 30 confirmed photographs.

3.5 Component 3 summary

Component 3 is owned by Vithanage P. V. P. M. (IT22062642). It addresses the third subproblem, deciding when and for how long each section of a shade house should

be watered, when its humidity tray should be filled, and when it should be fed. Vanda grow bare root, so the substrate moisture reading that ordinary automatic irrigation depends on does not exist, and because Vanda are watered daily the useful decision is not whether to water but when and for how long. The inputs are air temperature, humidity and light measured by an ESP32 sensor node in each section, together with the grower's settings for growth stage and tray amount. The outputs are a daily plan for each section, with a watering time between 06:00 and 09:00 and a duration of 30 to 120 seconds, humidity tray fills, feeds attached to a watering, and alarms, all carried out through a master controller that confirms every pour.

The component works as a sense, decide and act loop. A DHT22 and a BH1750 on each node report every 15 or 60 seconds, and a node keeps up to 8,640 readings in flash while the network is down. One master controller per house drives two 12 V pumps through an active low relay board, one for watering and one for the humidity tray, so that a tray fill can never wet the roots. In the cloud, the automation engine wakes every 60 seconds and does whatever is due (Figure 3.4). At 05:00 it plans every section from its dawn reading and the whole of yesterday's stored history, between 06:00 and 09:00 it waters each section at its own predicted minute, and every 15 minutes in daylight it decides whether each section's tray needs filling. A pour counts as done only when the master acknowledges it with the seconds the relay was actually on.

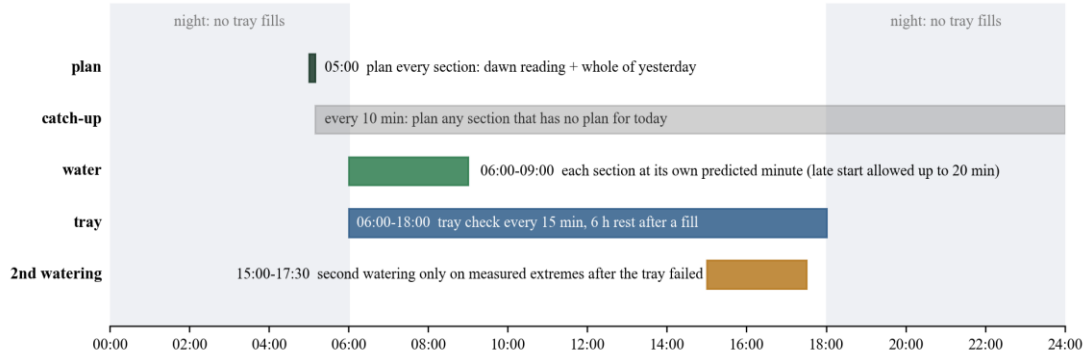


Figure 3.4: What Component 3's automation engine does over a day. Each section is planned at 05:00 and watered at its own predicted minute between 06:00 and 09:00, trays are checked every 15 minutes in daylight, and a second watering is allowed only on measured extremes after the tray has failed

Watering time and duration are predicted by two random forests [3] from ten features, namely the four dawn readings, three summaries of the previous day, the month, the growth stage and the section's light exposure. No record exists of when a skilled grower would have watered each section on each day, so the models were trained on 58,436 section-days built from eight years of ERA5 reanalysis [5] for five Sri Lankan sites, converted to shade house conditions, with targets generated by an expert watering rule from the previous day's evaporative demand. Trained on 2017 to 2022 and tested on 2023 and 2024, the time model is off by 8.42 minutes on average with a coefficient of determination of 0.954, and on weather from January 2025 to August 2026, downloaded after training, it is off by 8.41 minutes. The duration model is off by 2.54 seconds. Both errors sit at the noise built into the labels, 8.62 minutes and 2.39 seconds, which means the models reproduce the rule on years they never saw and cannot be better than it.

The humidity tray loop is the stronger result, because its target is the humidity series itself rather than a rule. Every 15 minutes in daylight a tray is filled when humidity is below the 60 to 80% band, or when a random forest classifier predicts from the section's own recent trajectory that it will fall below 60% within three hours. On 47,515 daylight hours of 2023 and 2024 that classifier reaches an area under the curve of 0.986, and at the deployed threshold of 0.70, chosen to favour precision, 91.5% of its early fills are followed by a real fall. The models decide when a tray is filled and the grower decides how much it pours, since a probe that measured the tray's level was removed after one read 0% for thirty hours on the farm.

Fertiliser is scheduled by growth stage, 30-10-10 during active growth and 10-30-20 during flowering, every 7 days at half strength and every 14 days at quarter strength from November to February. A dormant plant is never fed, which is a hard rule in the backend, and a due feed is attached to the section's next watering so that it always reaches wet roots. The two decision trees that encode this schedule score an F1 of 1.000 on held-out data, and that figure is not reported as a learned result, because their labels are a deterministic function of the growth stage and the days since the last feed. They are kept in classifier form only so that they can later be retrained on

recorded grower decisions. The dosing pump is not yet built, so the grower mixes the feed into the reservoir when the application says it is due.

The number of sensors a house keeps is measured rather than assumed. During a calibration period one node records in each section, and an eight step analysis measures how far the sections differ and how far one node's reading stays representative, then chooses which sections keep a sensor, never fewer than four. Sections left without a sensor are estimated by ordinary kriging [17] from their neighbours, labelled as estimated, and planned and watered like any other. On data with a known answer the analysis recovered an injected bias of +0.4 °C as +0.383 °C and a 5 m coverage radius as 5.03 m.

Component 3's contribution to the overall solution is that it acts on the shade house rather than advising the grower, and it applies the system's refusal principle to actuation. No plan, watering or tray fill is made from a reading more than two hours old, the master refuses a command more than 15 minutes old and caps every pour at 120 seconds, and a watering, tray fill or feed is recorded only when the hardware confirms it. It also supplies infrastructure the other three components run on. The account system built for it, with admin, operator and viewer roles, provides sign-in for the whole application, and the backend server set up for it hosts all four components' services. Its own decisions take no input from the other components, because the growth stage that selects the feed is set by the grower, a choice made at the planning stage so that watering can run on its own.

3.6 Component 4 summary

Component 4 is owned by Samadinee L. D. N. (IT22083500). It addresses the fourth subproblem: deciding which phenological stage a plant has reached, and how many days remain before it blooms. Its input for the first question is a plant photograph; for the second it is a photograph together with the temperature, humidity and light intensity recorded when the photograph was taken. Its outputs are one of six stages (vegetative, budding, pre-bloom, full bloom, wilting and seed formation) with a confidence, the two runner-up stages and the care protocol for that stage, and an estimate of the days until bloom together with the calendar date it implies.

Component 4 addresses a gap left by the two closest published systems. A 2025 multi-modal optical sensing study [12] monitors orchid growth through hyperspectral imaging, fluorescence spectroscopy and machine vision, and reports bloom prediction at a 21-day lead time, but it is built around Phalaenopsis. It does not model the growth habit that distinguishes Vanda, a single upward stem with aerial roots that can flower several times a year; it has no model for repeated flowering; and it does not monitor the plant after bloom. Its sensing also depends on hyperspectral and fluorescence equipment that a smallholder grower does not own. OrchiZen [13] classifies orchid growth from photographs with a YOLO detector and decision-tree and nearest-neighbour classifiers, but its stages (baby, child, young, mature and adult) describe the age of the plant rather than its flowering cycle, and they are used only to recommend fertiliser; it does not predict bloom. Component 4 therefore classifies the stages of the Vanda flowering cycle itself, from budding through full bloom and wilting to seed formation, from an ordinary phone photograph, and pairs the stage with a bloom-timing estimate from the same photograph and the shade-house readings.

The stage classifier is an EfficientNetB0 [1] pretrained on ImageNet, trained in two phases: first with the convolutional base frozen and a new pooled, regularised dense head, then with the last thirty base layers released at a tenth of the learning rate. The dataset contains 11,562 images taken on 35 capture dates. Two data defects were corrected before the reported run. Photographs saved in formats the training loader did not recognise were being silently skipped, and byte-identical photographs filed under two different stage folders were teaching the model contradictory labels; a cleaning pass converted the former and quarantined 780 files of the latter kind for review rather than deleting them. The images are divided into training, validation and test sets at 70, 15 and 15 per cent, stratified by stage. On the 1,768 held-out test images the classifier reaches 0.940 accuracy and 0.94 macro F1 against a majority-class baseline of 0.196.

The remaining error is concentrated in one place. Vegetative plants have the lowest recall, 0.83, and 32 of the 280 vegetative test images are classified as seed formation, the one confusion that exceeds ten per cent. Both are stages with no open flower, and

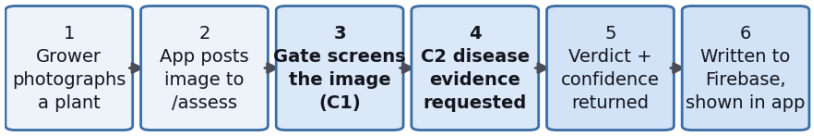
in both the frame is dominated by leaves and stem, so the confusion is the one a grower would expect. A second limit is the unit of grouping: the photographs carry no plant identifier, so two different photographs of the same plant can still fall on opposite sides of the split, and the 0.940 figure is best read as accuracy on new photographs rather than on new plants. Where a photograph contains several plants or several flower bunches, a second path locates each object with the open-vocabulary OWLv2 detector [11] and classifies each crop separately, rather than returning one stage blended across the frame. Unlike Component 1, the component has no novelty gate, so an image that is not an orchid still receives a stage.

Bloom timing uses a two-branch network: an EfficientNetB0 image branch reduced to a 32-value embedding is concatenated with a small dense branch over the three standardised ambient readings, and a single non-negative output gives days until bloom. The training record covers photographs taken on 12 dates between 12 July and 4 August 2026, a total of 12,260 images with a target between 0 and 23 days, divided at random into training, validation and test sets at 70, 15 and 15 per cent. On the 1,839 test images the regressor reaches a mean absolute error of 0.92 days, a root mean squared error of 1.41 days and a coefficient of determination of 0.938. Predicting the training mean for every test image gives a mean absolute error of 4.86 days, so the model reduces the naive error roughly fivefold.

Component 4's contribution to the overall solution is the timing information the other components act on. The stage determines the fertiliser formulation that Component 3 applies: the six stages map onto the Active, Flowering and Dormant states its planner takes as input, and the stage protocol switches from a balanced feed to a bloom formula at budding and back after wilting. The bloom estimate bounds the window in which Component 1's pollination assessment is useful. Both halves clear the baseline of their own split by a wide margin.

3.7 End to end user and data flow

Figure 3.3 traces one assessment from the grower's hand to the recorded verdict. The path is short by design: six steps, one network call, and no state held in the application that is not also written to the database.



Step 4 is the integration point the panel asked for: Component 1 does not re-implement disease detection.
It calls Component 2's classifier through LocalDiseaseProvider and folds the returned severity into its own trait vector.
If Component 2 is unavailable the provider reports zero confidence and C1 proceeds on image evidence alone, rather than silently treating "unknown" as "no disease".

Figure 3.4: End-to-end path of a single pollination assessment, showing the point at which Component 1 consumes Component 2's disease classifier rather than re-implementing it

The fourth step is where the four components stop being parallel. Component 1 needs to know whether the plant in front of it is diseased, which is precisely the question Component 2 exists to answer. Rather than train a second disease model on its own smaller dataset, Component 1 calls Component 2's classifier through a provider interface and folds the returned severity into its trait vector on a three-point scale. The interface is described in Section 4.2.

The failure behaviour of that call is as important as its success behaviour. If Component 2's model is unavailable, the provider reports zero confidence and Component 1 proceeds on image evidence alone. It does not treat an unidentifiable leaf as a healthy one. An earlier version of the provider checked only whether the model file existed and reported itself available whenever it did, which meant that on a server without the deep-learning runtime installed the system silently fell back to a colour heuristic while continuing to report that the trained classifier was in use. That defect is recorded here because the corrected check, which verifies that the runtime can actually be imported, is now the pattern used for every optional dependency in the backend.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

Integration was interface-first. The four components agreed the shape of the database and the signatures of the shared endpoints before any model was trained, which meant that each component could be developed against a stub of its neighbours and joined later without renegotiation. The alternative, integrating four finished components at the end, was rejected early; with four authors working to one deadline it concentrates all the risk into the period when there is least time to absorb it.

Components were joined in dependency order. Component 3's database schema was settled first, because the account system and the backend server that all four components use were built with it. Components 1 and 2 were then integrated as independent routers. Component 1 was integrated last, because it is the only component that consumes another component's model output at run time and therefore could not be finished until Component 2's classifier was stable.

The principal integration risk was that one component's change would silently break another's assumptions. Two measures addressed it. Interface ownership is explicit: the component that writes a database path owns its schema, and no other component may write there. And the integration of Component 2 into Component 1 was implemented as a provider with a declared contract rather than a direct import, so that Component 2's internals can change without Component 1 being rebuilt.

4.2 Interfaces data contracts and shared models

Table 4.1 documents the interfaces that cross a component boundary. Internal endpoints used by a single component are omitted; they are documented in the relevant individual thesis.

Table 4.1: Interfaces that cross a component boundary, with the error behaviour each consumer may rely on

Interface	Owner	Consumer	Contract and error behaviour
POST /api/v1/pollination/assess	C1	Mobile app	Multipart image plus optional traits. Returns verdict, confidence and trait provenance. Returns 422 with a structured body when the input gate refuses the image; the app renders this as a result, not an error
POST /api/v1/pollination/compatibility	C1	Mobile app	Two parent names, pod parent first, plus optional health assessments. Returns an evidence tier and the registered precedents behind it. Never returns a probability
/api/v1/pollination/log/*	C1	Mobile app	The grower's crossing logbook. Records an attempted cross with both parents photographed, advances it through the stages a capsule passes, and closes it at the gate where it stopped. Returns counts with their denominators and suppresses rates below a minimum sample size
LocalDiseaseProvider	C2	C1	In-process call returning condition and severity on a three-point scale. An unidentified or invalid image returns confidence 0.0, which is distinct from, and must never be collapsed into, a report of no disease
/tenants/{t}/farm/houses/{h}/sections/{s}	C3	Mobile app	Each farm's data sits under its own tenant and is served only to that farm's signed-in users. A section holds its latest reading, today's plan and its tray, fertiliser and control state, with reading history in its own branch and pump commands in the master's queue. A failed sensor channel is written as -999 and dropped, never replaced by a default, and no decision is made from a reading more than two hours old
GET /api/v2/care/model-info	C3	App, examiners	Reports the metrics and baselines of the deployed models, so the figures shown in the application are read from the models rather than transcribed
/api/v2/accounts and Firebase Authentication	C3	All four components, mobile app	Sign-in for the whole application. Each farm is a separate tenant whose admin creates its accounts, and public sign-up is off. Component 3's endpoints check the user's role, admin, operator or viewer, on every action that changes anything, and a viewer can never start a pump

One shared convention deserves explicit statement because it was the source of a defect. Across the whole system, the absence of a measurement and a measurement of absence are represented differently. A trait the grower did not supply is omitted from the request entirely; a trait measured and found absent is sent with its value. An earlier version of the application sent every unanswered field as the string "unknown", which the backend could not distinguish from a deliberate answer, and the image-derived value was consequently overwritten every time.

4.3 Shared nonfunctional requirements

Performance targets were set by the interaction rather than by the hardware. A grower standing in a shade house will wait a few seconds for a verdict, so the assessment path is budgeted at under five seconds end to end on a mid-range Android device over mobile data. Restructuring the database to move history out of the section subtree reduced the dashboard payload from roughly one megabyte to seven and a half kilobytes and the overview call from 4.1 to 2.0 seconds, which was the single largest contributor to meeting that budget.

Security is modest but deliberate. Write endpoints require an API key held in a configuration file that is excluded from version control. This is not a theoretical control: the repository is public and the first key was committed and published before the omission was noticed, after which it was rotated and the exclusion added. The incident is recorded here rather than quietly fixed because it is the reason the team now treats secret handling as a shared non-functional requirement rather than each component's own business.

Component 3 adds stronger controls, because its endpoints can start a pump. Each of its 69 endpoints requires a signed-in user and serves only that user's farm, every endpoint that changes anything requires a sufficient role, and all traffic is encrypted with Transport Layer Security (TLS). A database write that fails is reported to the caller as a failure, never as a success. The database security rules still have to be closed before the system is sold.

Reliability rests on the principle that no component may fail silently. Every optional dependency is checked by attempting to import it, not by looking for a file; every failed sensor reading is dropped and counted rather than replaced by a default; and every verdict carries the provenance of the traits behind it, so a grower can see which values were measured and which were assumed. Maintainability is served by keeping the backend address and the API key in one configuration module each, after an earlier arrangement that copied the address into ten files made a change of network a ten-file edit with one file routinely missed.

For accessibility, Component 3 offers a simple Today screen with large text, plain words and one button beside the expert dashboard, and its alarms appear full screen over the lock screen at alarm volume, repeating every five minutes up to six times until someone acts. For electrical safety its relay boards are opto-isolated and active low with their pins driven off before use, every pump carries a flyback diode, and the 120 second pour limit and the refusal of stale commands are enforced in both the firmware and the backend.

4.4 Integrated workflow and exception handling

In normal operation the grower opens the application, selects a plant, and either photographs it or reads the current sensor state. The relevant component returns a verdict, the verdict is written to the section's subtree, and the application displays it with its confidence and the evidence behind it. For Component 3 the loop closes without the grower: the automation engine plans at dawn, checks trays every fifteen minutes and evaluates watering every minute, and either acts or raises an alarm depending on whether automatic mode is on.

Three exception paths matter at the system level. The first is a refused input: when Component 1's gate declines an image, the backend returns a structured 422 and the application presents the refusal as its own result with an explanation, rather than as a network error. The second is a missing dependency: when a component's model cannot be loaded, the affected endpoint reports that it is unavailable instead of falling back to a heuristic while claiming to be the model. The third is a failed sensor or device. A failed reading is dropped rather than replaced by a default, no watering or tray decision is made from a reading more than two hours old, and a sensor failing three readings in a row, a node silent for twenty minutes or a pour the master never carries out each raises an alarm. On the deployed system a failed temperature sensor raised its alarm 3 minutes 17 seconds after its first failed reading.

Automatic mode deserves a note because its failure semantics are frequently misunderstood. Switching automation off does not stop the system observing or deciding. It changes only whose hands execute the decision: with automation on the

system acts and reports what it did, and with automation off it raises a push notification telling the grower what needs doing. The brain stays on in both cases.

4.5 Team contribution and dependency matrix

Table 4.2 records the division of work. It is organised by task type rather than by chapter so that shared work is visible as shared rather than being attributed to whoever wrote it up.

Table 4.2: Division of research, development and integration work across the four component owners. Entries marked for completion are to be filled by the owner concerned

Area	C1 owner	C2 owner	C3 owner	C4 owner
Research and dataset assembly	Lead - 1,190 photographs, 28 tagged plants, RHS record	Lead - 667 field photographs, 427 severity labels	Lead - farm visit, ERA5 2017 to 2024 for five sites, expert rule labels	Lead - 11,562 stage-labelled images; bloom record over 12 dates
Model development	Lead - gate, flower filter, suitability, compatibility, crossing logbook	Lead - disease, severity, input gate, treatment rules	Lead - watering, tray and feed models, placement analysis, kriging	Lead - stage classifier, bloom regressor, dataset cleaning pass
Backend routers	Lead - pollination router, logbook endpoints, disease provider	Lead - disease router, inference used by C1	Lead - automation engine, device health, alarms, accounts and roles	Lead - growth and bloom routers, per-object detection path
Mobile screens	Lead - pollination, varieties, cross check, crossing logbook	Lead - detection, history, contribute	Lead - farm, Today, Run, Alarm, calibration and team screens	Lead - growth stage and bloom prediction screen
Shared infrastructure	Contributed - input validation pattern, secret handling	Contributed - cross-component import fix	Lead - account system, server for all four, Firebase schema, node hardware	Contributed
Integration and test evidence	Lead - gate validation set, grouped CV protocol	Lead - leakage check, held-out evaluation	Lead - digital twin, 19 scenario end-to-end campaign, farm deployment	Lead - grouped split manifests

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The system is built on React Native with Expo for the application, FastAPI on Python for the backend, and Firebase Realtime Database for shared state. Models are TensorFlow and Keras for the three deep-learning components and scikit-learn with XGBoost for the tabular and novelty-detection work. Sensor nodes are ESP32 microcontrollers carrying a DHT22 temperature and humidity sensor and a BH1750 light sensor, and in each house one node with a four channel relay acts as the master controller, driving a watering pump and a humidity tray pump. Component 3's models are random forests and decision trees in scikit-learn, its spatial estimation uses PyKrige and PySensors, and its alarms are sent through Firebase Cloud Messaging.

Two choices were made jointly and are worth justifying. Firebase Realtime Database was chosen over a relational store because three of the four components publish a stream of readings or verdicts keyed by device, and because the microcontrollers can write to it directly without an intermediate service. A single FastAPI process hosting four routers was chosen over four services because the components share pre-processing, and running them in one process is what guarantees that an image is segmented identically regardless of which component received it.

Dependency management became a shared concern rather than a per-component one after a deployment failure traced to version skew: models pickled against a newer numerical library could not be loaded by the server, and a gradient boosting library present in development was absent in production. The pinned requirements file is now shared, models are refitted in an environment matching the server's pins before release, and a trained model is unwrapped from any helper class defined in a training script before it is serialised, so that the serialised file does not depend on code that is not deployed.

5.2 Development and integration process

Work was organised in sprints against a shared board, with each component owner responsible for their own backlog and the integration tasks held jointly. The repository is a single Git repository with one branch per component feature and review before merge to the main branch. Keeping all four components in one repository was a deliberate choice: it makes a change that breaks an interface visible immediately rather than at integration time.

The practice that proved most valuable was requiring a measured comparison rather than an asserted choice. Where a component selected a model, it was expected to report the alternatives it was selected over, on the same split. That requirement is why Section 6.3 is able to show, for Component 1, that two of the four candidate classifiers do not reach the majority-class baseline at all, which is considerably more informative than reporting the winner alone.

5.3 Deployment and operational configuration

The backend is deployed to a virtual private server reachable over a domain name, with the application pointing at it through a single configuration module. Models are loaded at process start and held in memory; the model registry reports which are loaded, so a partially provisioned server is visible rather than silent. The database needs no deployment step, since the sensor nodes write directly to Firebase.

The operational lesson of this project concerns environment parity rather than deployment mechanics. The first deployment to a shared server failed not because the code was wrong but because the environment differed from the one the models had been fitted in. The remedy adopted, refitting the serialised models inside an environment that matches the server's pinned versions and verifying that each loads before release, is now a release step rather than a debugging technique.

Component 3 adds its own operational arrangements. The server is an Ubuntu 24.04 virtual private server with 4 cores and 8 GB of memory, costing USD 6.60 per month, behind the Caddy web server, which terminates TLS. The backend is deployed by hand, by pulling the main branch and restarting the service, and never

between 06:00 and 09:00, so that a restart cannot fall in the watering window. Node firmware is compiled with arduino-cli and flashed over USB, after which a node registers its Media Access Control (MAC) address and fetches its house and section assignment, and an unassigned node uploads nothing. Monitoring is automatic. The automation engine writes a heartbeat on every pass, each node sends a heartbeat of its own, and the device health checks raise an alarm when a sensor fails, a node falls silent or the master does not carry out a pour.

5.4 Evidence of team execution

Execution evidence falls into three kinds. The repository history records what was built and when, including the sequence of corrections described throughout this report. The simulation and failure-case suites record what was tested. And the research diaries record the decisions that were taken and reversed, of which the reformulation of the watering question and the regrouping of the pollination validation are the two with the largest effect on the final system.

The repository is the primary record. Between 19 April and 11 October 2026 it accumulated 353 commits, of which 277 carry changes and 76 are merges, across 74 pull requests. Every component was developed on its own long-lived branch and every change reached main through a pull request rather than a direct push, so the review history is itself part of the record. Figure 5.1 shows the distribution over the project period.

Automated verification is the second record. The backend carries 327 test functions across 29 suites, run by GitHub Actions on every push and pull request, and a test that fails blocks the merge. These are not only unit tests of behaviour: several assert properties of the system that no behavioural test would notice, the clearest being the check that no backend file reaches the shared database outside the four agreed helpers. That one rejected a change during development whose call would otherwise have succeeded silently against the wrong path.

The third record is the written one. Design decisions that spanned components were specified before they were built and the specifications were kept: five staged plans and one design document cover the multi-tenant account work alone, and the breeder

interview that the pollination knowledge base rests on is held in full under docs/research/. Where a decision was reversed, the reversal is recorded with it. The two with the largest effect on the final system are the reformulation of the watering question and the regrouping of the pollination validation, both described in the chapters that report them.

Per-member contribution is set out in the matrix of Section 4.5 rather than as a commit count. Commit frequency measures activity, not contribution, and the two diverge sharply here: work carried out in notebooks and committed as a finished model artefact appears as a single commit beside the same quantity of work done incrementally. The matrix names what each member led, which is the claim this report actually makes.

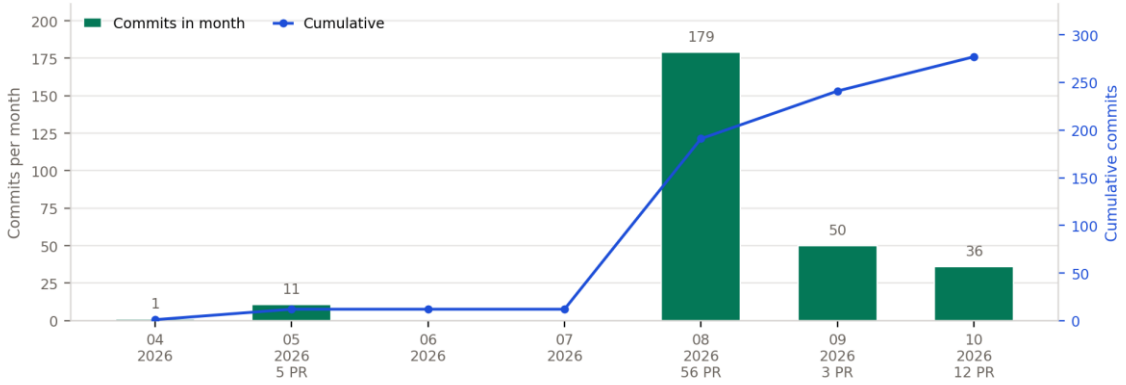


Figure 5.1: Commits and merged pull requests per month over the project period. The gap across June and July is the inter-semester break; the August peak is the integration push that brought the four components onto one backend.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation framework was fixed before results were collected, and rests on three rules applied to all four components. Every reported figure is accompanied by the majority-class baseline, or by the equivalent naive error, of the same split. Every split reflects the deployment condition rather than the convenience of the data. And a result that falls below its baseline is reported as such rather than omitted.

The second rule has a concrete meaning that differs by component. For Component 1 it means grouping by physical plant, because 357 photographs come from 28 plants and an image-level split lets a classifier identify the plant instead of its condition. For Component 3 it means splitting out of time, training on earlier years and testing on later ones, and additionally testing on two years downloaded after training had finished. For Components 2 and 4 it means a held-out test set that is not the validation set used for model selection.

The group-level success criteria follow the objectives of Section 2.4. The system is judged successful on G1 if each component exceeds its baseline on a deployment-realistic split; on G2 if a verdict produced by one component is demonstrably consumed by another at run time; on G3 if every figure in this report carries its baseline, including the unfavourable ones; on G4 if the input gate refuses invalid inputs at a measured rate on data never used for fitting; and on G5 if at least one leakage or validity defect was detected and corrected by the protocol itself.

6.2 Integration and end to end testing

Integration testing covered the interfaces of Table 4.1 and the exception paths of Section 4.4. Interface tests confirm that a refused image produces a structured 422 rather than a generic failure, that an unavailable disease model yields zero confidence rather than a report of health, and that a trait omitted by the application is not overwritten by a default. End-to-end tests exercise the full path from photograph to recorded verdict against a running backend.

System-level behaviour under hardware conditions was tested using a simulator that writes to the real database on the paths the microcontrollers use, polls for commands and executes them, and models tray evaporation and humidity. It found defects that unit tests had not. The automation loop diagnosed empty trays correctly and never refilled them, the tray's rest period was timed on the server's clock rather than the device's, several screens never refreshed so data froze until the user navigated away and back, and the first planner split two shaded spots differing by 0.2% per hour. Component 3 was then run through an end-to-end campaign of nineteen scenarios on the deployed backend and the real application, against a simulated house whose virtual boards follow the firmware's protocol, covering calibration, manual watering and stopping, tray fills, scheduled watering with automatic care on and off, feeds, repeated alarms, every device health fault, a Wi-Fi outage and an offline master. It found 25 defects, one critical, nine high, six medium and nine low, and every critical, high and medium defect was fixed, after which all nineteen scenarios passed. The most severe left sections without their own sensor unplanned and unwatered.

Component 3's backend has 368 automated tests in 30 files, with one regression test for each defect the campaign found. In the last full run 336 passed and 31 were skipped, the skipped ones being integration tests that need the live service and its key, and one failed only when every file runs in one process, a test harness defect recorded as open. The 304 tests of the continuous integration job all passed. Two of the tests check that every control the application shows is gated by the permission the server enforces. Measured on the deployed system, a manual watering command reached the master 3 seconds after issue, an unacknowledged alarm repeated after 5 minutes 8 seconds, and 18 readings buffered during a Wi-Fi outage were uploaded in time order on its return.

6.3 Overall results

Table 6.1 collects the headline result of each component beside the baseline that makes it interpretable. The components are not directly comparable with one another, since two are classification tasks, one is a regression over time and one combines both; the table is a summary of evidence rather than a league table.

Table 6.1: Headline results for the integrated system, each stated beside the baseline or naive error of the same split

Component	Task	Result	Reference value	Evaluation split
C1	Pollination suitability, 3 classes	Accuracy 0.717, macro F1 0.715	Majority baseline 0.639	Grouped by plant (StratifiedGroupKFold) , n = 357
C1	Input gate, non-orchid refusal	91.9% refused, AUC 0.921	309 of 309 non-plant images refused	Held-out validation set, n = 455
C2	Disease classification, 3 classes	Accuracy 0.806, macro F1 0.778	Majority baseline 0.418	Held-out test, n = 67 leaves
C2	Severity grading, 3 grades	Macro F1 0.450	Majority baseline 0.395	Held-out test, n = 43 diseased leaves
C3	Watering time, 06:00 to 09:00	MAE 8.42 min, R^2 0.954, and on unseen weather 8.41 min, R^2 0.952	Label noise floor 8.62 min	Out-of-time, train 2017-22, test 2023-24, n = 14,620, and unseen Jan 2025 to Aug 2026, n = 2,364
C3	Humidity below 60% within 3 hours	AUC 0.986, precision 0.915 and recall 0.648 at threshold 0.70	No-skill AUC 0.500, falls in 9.1% of daylight hours	Out-of-time, 2023-24, n = 47,515 daylight hours
C4	Growth stage, 6 classes	Accuracy 0.940, macro F1 0.94	Majority baseline 0.196	Held-out test, n = 1,768
C4	Bloom timing	MAE 0.92 days	R^2 0.938; mean-prediction MAE 4.86 days	Held-out test, n = 1839

Figure 6.1 presents the three classification accuracies against their baselines. Component 3 is deliberately absent from it. Its watering model is a regression reported in minutes, and its humidity classifier is scored by area under the curve because falls occur in only 9.1% of daylight hours, so a classifier that never predicted a fall would already be 90.9% accurate. Plotting either on an accuracy axis would invite a comparison that is not meaningful.

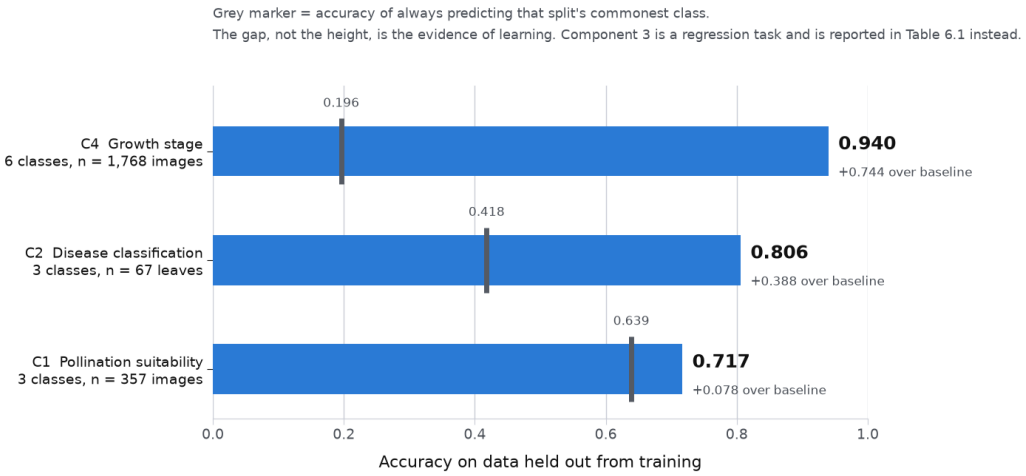


Figure 6.1: Classification accuracy of three components against the majority-class baseline of the same held-out split. The distance between the bar and the marker is the evidence of learning

The figure makes one feature of the results immediately visible. Component 2's margin over its baseline is large and Component 4's is larger still in proportional terms, while Component 1 clears its baseline by 0.078. That narrow margin is a consequence of the class distribution: 228 of the 357 images, drawn from 17 of the 28 plants, are labelled Suitable, so a classifier that answers Suitable every time is already right 63.9% of the time and the available headroom is correspondingly small.

Because that margin is narrow, Component 1's model selection is reported in full rather than by its winner alone. Figure 6.2 shows all four candidates scored on the same grouped splits.

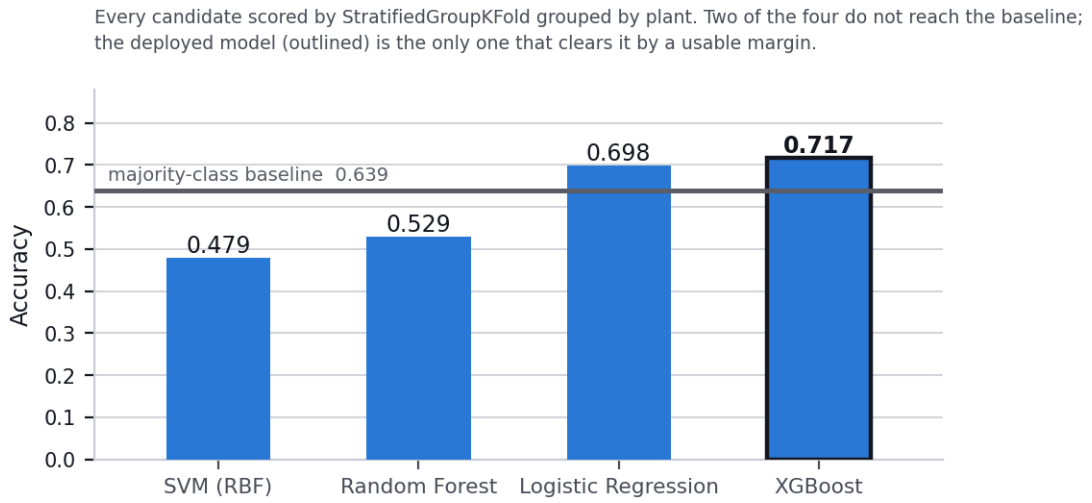


Figure 6.2: Component 1 model selection. All four candidates scored under StratifiedGroupKFold grouped by plant; the support vector machine and the random forest fall below the majority-class baseline

Two of the four candidates perform worse than always answering Suitable. This is a more useful result than it first appears: it establishes that the margin achieved by the selected model is a property of that model rather than of the task being easy, and it is the kind of comparison that a report quoting only the winning figure would conceal.

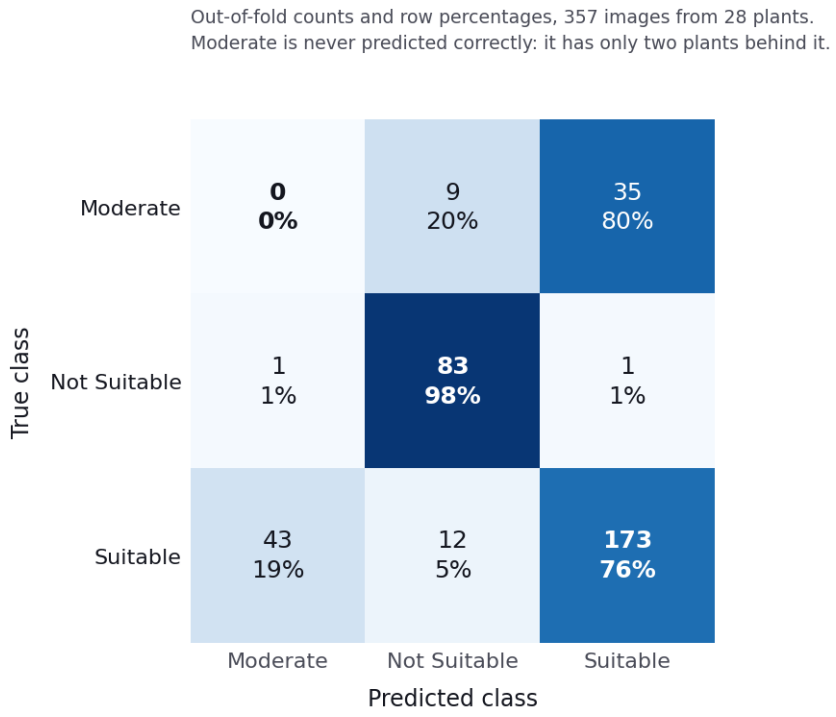


Figure 6.3: Out-of-fold confusion matrix for Component 1's deployed classifier, with counts and row percentages

The confusion matrix in Figure 6.3 exposes the component's real limitation. Not Suitable is recovered almost perfectly, at 98%, which is the clinically important direction: the system rarely tells a grower to breed from a plant that should not be bred from. Suitable is recovered at 76%. The Moderate class is never predicted correctly at all. There are only two plants behind that class, which is too few for the grouped cross-validation to learn or to validate it, and its per-class scores are therefore reported as indicative only. Enlarging it is the first item of future work in Section 8.3.

Table 6.2 reports the input gate, which is evaluated separately because it is answering a different question: not how accurate the verdict is, but whether a verdict should have been issued at all. The validation set was assembled from sources never used in fitting.

Table 6.2: Component 1 input gate measured on a held-out validation set of 455 non-orchid and 88 unseen orchid photographs. Grouped cross-validation on the project's own photographs gives a false-refusal rate of 2.2%

Category	n	Accepted	Interpretation
Cars, documents, food, laptops, people, rooms, screens	309	0%	No non-plant image was accepted

Category	n	Accepted	Interpretation
Houseplants	48	15%	Foliage is the hardest negative class
Ferns	51	6%	Refused despite similar leaf geometry
Other flowers	45	60%	Passed stage 1. Stage 2 refuses 78% of these
Vanda, unseen web photographs	43	65%	Accepted. The rest are false refusals
Other orchid genera, web	45	67%	Accepted as orchids

The gate is strongest where the consequence of error is worst and weakest where it is mildest. Nothing that is not a plant is ever accepted. Other flowers pass the first stage at 60%, which is why the second stage exists; it refuses 78% of roses, tulips and sunflowers, with an area under the curve of 0.868 measured against a third-party flower dataset. The cost of this conservatism is that roughly a third of genuine orchid photographs from unfamiliar sources are refused. For this application that trade is the right way round: a refusal costs the grower a retake, whereas a confident verdict on the wrong species costs a breeding season.

6.4 Stakeholder and user validation

Validation by stakeholders was carried out in two forms. Growers were shown the application and asked to complete representative tasks, and their reactions were recorded against the task rather than against the interface. The breeder consulted on crossing reviewed Component 1's compatibility output and confirmed the decision not to report a success percentage, on the grounds that the registered record does not contain the failed attempts that such a figure would require.

For Component 3, The system was evaluated with the owner of the orchid farm after testing the hardware and application on site for two weeks. Following this trial, an interview was conducted to get direct feedback on daily use, time savings, and practicality. The grower explained that watering the shade houses by hand normally takes about three and a half to four hours every day. Managing watering for the sections from one system cut out this daily manual work and saved significant time. He noted that growers normally guess moisture and heat just by looking around, which is not consistent, so having exact numbers and automatic control makes daily

care much easier. He also welcomed the automated fertilization feature, noting that delivering feeds with the watering saves extra labour and ensures plants receive nutrients regularly. The grower also highlighted the value of keeping a continuous climate history. General weather forecasts do not show what happens inside a shade house, and growers do not write down daily numbers. Having stored records of past temperatures, humidity drops, and waterings allows a grower to look back. Such as during hotter months and see how the shade house actually behaved. Finally, the grower praised the hardware design, noting that the IoT device is low cost and does not use expensive or complicated gadgets. Because small commercial growers cannot invest in costly equipment, keeping the setup simple, cheap, and effective makes it practical for real farm adoption.

6.5 Achievement of common objectives

Table 6.3 maps evidence to each of the five group objectives stated in Section 2.4 and states whether it was achieved in full, in part, or not at all.

Table 6.3: Evidence for each group objective and the outcome claimed

Objective	Evidence	Outcome
G1 Four components, each evaluated on held-out data	Table 6.1; all four report a held-out figure and six of eight exceed their reference value	Achieved in part. Severity grading does not reach their targets and is reported as such
G2 Joined through one backend, database and application	Figure 3.1; Table 4.1; Component 1 consumes Component 2's classifier at run time through LocalDiseaseProvider	Achieved
G3 Every claim stated beside a baseline, including unfavourable ones	Table 6.1 and Figures 6.1 to 6.2; the severity-grading shortfall and the two sub-baseline classifiers are reported	Achieved
G4 System refuses inputs it cannot interpret	Table 6.2; no non-plant image accepted, AUC 0.921 on a held-out set	Achieved
G5 Splits reflect deployment; protocol detects its own defects	Section 2.2; the image-level split that reported perfect accuracy was detected and corrected to a grouped split reporting 0.717	Achieved

The honest summary is four achieved and one partly achieved. G1 is reported as partial because one of the eight measured tasks does not clear the reference value of

its own split: severity grading reaches 0.450 macro F1, It is retained in the system and reported, rather than removed to improve the summary.

6.6 Integrated limitations and threats

The most serious system-level limitation is sample size in the pollination component. Twenty-eight plants is the true sample, not 357 images, and no amount of photography of the same plants will change that. The Moderate class, with two plants, cannot be validated at all. Any claim made about Component 1's generalisation should be read with that number in mind.

A second limitation is that the power use of a node has not been measured, the first carrier board did not connect the ESP32 to the sensors so the farm nodes are wired by hand, and firmware 2.8, which lets a master water from its own copy of the day's plan when the cloud is unreachable, has been tested off the board but not yet flashed to the farm nodes

A third concerns the training of the deep-learning components on datasets whose photographic conditions differ from a Sri Lankan shade house. The input gate mitigates this for Component 1 by refusing images that fall outside the trained distribution, and the confidence damping reduces the strength of verdicts near the boundary, but neither mechanism converts an out-of-domain image into a reliable one. The flower-detection work provides the clearest illustration: a model trained on clean cutouts reached 0.999 cross-validated accuracy and then classified zero of 38 real non-flowering plant photographs correctly. That result is retained in the record because it is the sharpest available evidence that cross-validated accuracy on a convenient dataset predicts very little.

The principal threat to validity in the integrated evaluation is that the four components were evaluated separately and then composed. The report states what each component achieves on its own data; it does not yet state the accuracy of the end-to-end recommendation, which would require a labelled set of complete cases that the team does not have. Section 8.3 names this as the most valuable piece of outstanding work.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The value offered is the conversion of four experience-dependent judgements into recorded, measured recommendations available on a phone the grower already owns. For a smallholder the benefit is concentrated at the two points where error is expensive and late to appear: a disease caught at the right bench rather than the wrong week, and a breeding attempt made on a plant fit to carry it. For a nursery running a breeding programme, the compatibility record is the more valuable half, because it makes an institutional memory out of knowledge that currently leaves with the person who holds it.

For Component 3 the value lies in daily labour and record keeping. A grower who now waters every house by hand each morning gets each section watered at its own time and for its own length, humidity raised without wetting the roots, feeds given only with water and never to a dormant plant, and a record of every pour confirmed by the controller that ran it. Because the calibration decides how many sensors a house keeps, the grower buys only the nodes a house needs, and the owner can give workers operator or viewer access.

The beneficiary least often named is the new entrant. The barrier to growing Vanda commercially is not capital, since the plants and the shade house are modest investments by agricultural standards; it is the several seasons of losses that teach a grower to read a plant. A system that shortens that period lowers a real barrier to a high-value crop, which is the mechanism behind the Goal 8 claim in Section 1.3.

7.2 Market alternatives and differentiation

Three alternatives exist today. General-purpose plant identification applications are free and widely installed, but they identify species and offer generic care text; they do not assess the condition of a particular plant against a particular decision, and they answer confidently regardless of what they are shown. Commercial greenhouse

automation addresses irrigation and climate competently but is priced for industrial glasshouses and has nothing to say about disease or breeding. Agricultural extension services provide exactly the expertise required, and are the real substitute, but they are thinly spread and cannot be consulted at the moment a decision is taken.

The differentiation claimed is therefore not accuracy in the abstract. It is that the system covers four decisions a Vanda grower actually faces, in one application, at smallholder scale, and declines to answer when the evidence does not support an answer. The refusal behaviour is a differentiator rather than a limitation: it is the property that distinguishes the system from a general identification application that will confidently classify a photograph of a desk.

For irrigation specifically, generic controllers add a soil moisture threshold to a timer [18], which a bare root Vanda offers nothing to measure, and the closest integrated orchid platform, OrchiZen [13], advises the grower and leaves the action to the grower. Component 3 sits between the two. It acts, confirms what it did, plans each section separately and is built for epiphytes from commodity parts. Against established products its weaknesses are a short field record and its dependence on Wi-Fi and the cloud service.

7.3 Business model and feasibility

Technical feasibility is established: the system runs on a modest virtual private server, and the hardware for one sensor node costs approximately LKR 5,550 for the core sensing and camera unit, with the full development setup having cost LKR 7,550 including accessories. The marginal cost of an additional user is the server capacity to serve them, since the models are already trained and the database tier is shared.

The model that fits the beneficiary described in Section 7.1 is a free application with optional paid hardware. Photograph-based assessment requires no sensor node and can be offered without charge, which matters because the smallholders who would benefit most are the least able to pay in advance for software whose value they have not yet seen. The irrigation component requires hardware and is where a sustainable revenue line naturally sits.

Component 3's hardware is costed a sensor node, an ESP32 with a DHT22 and a BH1750, costs LKR 2,140 in parts, and one is needed for each section the calibration decides to keep. A house adds a master controller, which is a node with a four channel relay module at LKR 250, and two 12 V pumps at LKR 500 each, while tubing, trays and the 12 V supply. Three revenue sources fit this component. They are a hardware kit sold per house, a subscription per farm that pays for hosting, alarms, model updates and support and includes the farm's admin account, and calibration as a service, in which an installer calibrates a house with a set of nodes, leaves only the nodes it needs and reuses the rest, which lowers the upfront cost. One server at USD 6.60 per month serves every farm as a separate tenant, although its capacity for many farms has not been measured.

7.4 Adoption intellectual property and compliance

The principal adoption barrier is trust rather than cost or usability. A grower who has read plants for twenty years has little reason to defer to an application, and the correct response is not to argue but to make the system's reasoning inspectable: every verdict carries its confidence and the provenance of the traits behind it, and the system says when it does not know. The second barrier is connectivity, which is intermittent in rural areas and which the current design handles only partially. In Component 3 a node keeps its readings in flash while the Wi-Fi is down and uploads them in order on its return, and the offline fallback in firmware 2.8 lets a master water from its own copy of the day's plan, although that version is not yet deployed.

On intellectual property, the registered hybrid record used by Component 1 is third-party data used in a small, attributed, non-commercial research subset, and any commercial deployment would require the rights position to be settled with the publisher rather than assumed. Photographs of growers' plants were collected with permission. The repository is public, which is a deliberate choice for an academic project and the reason the secret-handling practice described in Section 4.3 exists.

On data protection, the system stores plant and equipment records and the account details of each farm's own users, and no other personal data. Grower accounts already exist. The account system built for Component 3 provides sign-in for the

whole application, holds each farm's data under its own tenant and checks each user's role, admin, operator or viewer, on every action that changes the farm, and public sign-up is off.

7.5 Sustainable Development Goal contribution

The plausible contribution to target 12.2 rests on a mechanism that is implemented and measurable. Irrigation is scheduled for each section from the drying power of the air rather than from a fixed clock, and fertiliser is withheld from dormant plants by a hard rule in the backend. Yesterday's mean vapour pressure deficit carries 0.949 of the importance in the watering time model and 0.888 in the duration model, but because the targets were generated by an expert rule built on evaporative demand, that figure shows the model learned the rule rather than proving the mechanism. The same mechanism bears on Goal 6 through target 6.4 on water use efficiency, since each section is watered for its own conditions instead of a whole house on one routine. What has not been measured is the resulting reduction in water and nutrient applied across a season in a working shade house, and until it has been, the contribution should be described as a plausible pathway rather than a demonstrated saving.

Two possible negative effects deserve naming. A system that recommends pollination attempts could increase the total number of attempts and therefore the resources consumed by unsuccessful ones, if its recommendations are weaker than a skilled grower's own judgement. And an automated irrigation system that fails in the open position wastes considerably more water than a manual one ever would, which is why the automation engine's failure paths were tested before its success paths.

Sustaining the benefit requires the models to be retrained as conditions change. The system is designed for this: the database accumulates the readings and verdicts that retraining would use. Whether that retraining actually happens after the project ends is an institutional question rather than a technical one, and the team does not claim to have solved it.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The common problem was that four recurring Vanda husbandry decisions are made by unaided visual judgement, inconsistently and without record. The integrated system answers that problem directly for three of the four decisions and partially for the fourth. Disease classification, growth-stage judgement and irrigation timing are each delivered by a model that clears the baseline of its own held-out split by a usable margin, and all three are available to a grower through one application with one record per plant.

The fourth decision, breeding, is answered with a deliberately narrower claim. Plant readiness is predicted at 0.717 against a 0.639 baseline, which is a real but modest margin on 28 plants, and cross compatibility is reported as an evidence tier rather than a probability because the registered record contains no failed attempts from which a probability could be computed. The team regards stating that limit as part of answering the problem rather than as a failure to answer it: a fabricated success percentage would have been easy to produce and would have been worse than the tier it replaced.

Judged against the five objectives, the project achieved four and partly achieved one. The partial achievement is sample size and task difficulty in two specific predictions, both of which are reported with their shortfall visible.

8.2 Principal integrated contribution

The contribution that exceeds the sum of the four components is the evaluation discipline and the input-validation behaviour, both of which began in one component and became properties of the system.

The evaluation discipline is the requirement that every figure appear beside the baseline of its own split and that every split reflect deployment. Its value is demonstrated rather than asserted: applied to Component 1 it converted a reported perfect accuracy into a measured 0.717 by revealing that the original split had placed

photographs of the same plant on both sides. A protocol that detects a defect in the team's own favour is stronger evidence of its worth than one that merely produces good numbers.

The input-validation behaviour is the two-stage gate, built after a classifier returned a confident suitability verdict for a photograph of a laptop. The general lesson, that a classifier trained only on positive examples will extrapolate confidently on anything it is shown, applies to all four components and is now the architectural pattern for every optional model and dependency in the backend. The team considers the system's willingness to refuse a question its most transferable result.

8.3 Future directions

Four pieces of work are prioritised by the strength of evidence they would add rather than by effort.

First, an end-to-end evaluation on labelled complete cases. The report currently states what each component achieves on its own data and not what the composed recommendation achieves, which is the honest gap identified in Section 6.6 and the single most valuable addition available.

Second, the data the grower logbook was built to collect. The instrument itself has been implemented: growers record an attempted cross with both parents photographed, the covariates that distinguish a genetically incompatible pairing from a mistimed or poorly executed one, and the stage at which the attempt stopped. The three outcomes are kept apart because they fail for different reasons. A failure to set a pod is usually technique, a capsule that aborts is usually pod-parent health, and a failure to germinate is where genetic incompatibility actually appears, and a record whose outcome is unknown is counted separately from the failures rather than with them. What remains is a season of entries, which no amount of engineering supplies: a Vanda capsule takes about nine months, so the outcome that would validate Component 1's own readiness model cannot be observed inside the period of this project. Until enough attempts have accumulated the record reports counts with their denominators and no rates.

Third, enlargement of the smallest classes, specifically the two plants behind Component 1's Moderate class and the severity grades that hold Component 2 to 0.450 macro F1. Both limits are sample size rather than method, and both are addressable by photography over a further season.

The fourth task is growers' corrections to the plan and the condition of the plants should be recorded over a season, and the watering models retrained on them. A current sensor in the pump supply would let the system notice a pump that is commanded but draws no current. Finally, delivery at each outlet, the nodes' power use and real water savings still have to be measured, and the carrier board remade with its tracks checked against the firmware's pin map.

REFERENCES

[1] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. 36th Int. Conf. Machine Learning (ICML), Long Beach, CA, USA, 2019, pp. 6105-6114.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Las Vegas, NV, USA, 2016, pp. 770-778.

[3] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5-32, 2001.

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, San Francisco, CA, USA, 2016, pp. 785-794.

[5] H. Hersbach et al., "The ERA5 global reanalysis," Quarterly Journal of the Royal Meteorological Society, vol. 146, no. 730, pp. 1999-2049, 2020.

[6] C. Rother, V. Kolmogorov, and A. Blake, "GrabCut: Interactive foreground extraction using iterated graph cuts," ACM Transactions on Graphics, vol. 23, no. 3, pp. 309-314, 2004.

[7] Royal Horticultural Society, The International Orchid Register. London, UK: Royal Horticultural Society. [Online]. Available: https://apps.rhs.org.uk/horticulturaldatabase/orchidregister/orchidregister.asp

[8] M.-E. Nilsback and A. Zisserman, "Automated flower classification over a large number of classes," in Proc. 6th Indian Conf. Computer Vision, Graphics and Image Processing, Bhubaneswar, India, 2008, pp. 722-729.

[9] D. M. Woebbecke, G. E. Meyer, K. Von Bargen, and D. A. Mortensen, "Color indices for weed identification under various soil, residue, and lighting conditions," Transactions of the ASAE, vol. 38, no. 1, pp. 259-269, 1995.

[10] United Nations, Transforming Our World: The 2030 Agenda for Sustainable Development. New York, NY, USA: United Nations, 2015.

[11] M. Minderer, A. Gritsenko, and N. Houlsby, "Scaling open-vocabulary object detection," in Advances in Neural Information Processing Systems 36 (NeurIPS), New Orleans, LA, USA, 2023.

[12] A. Sungheetha, R. Rajesh Sharma, B. P. Sreeja, S. Mahapatra, R. Thakur, and R. C. Karpagalakshmi, "Multi-modal optical sensing fusion for quantitative monitoring of orchid parameters: Analysis of commercial cultivation in Asia," in Proc. 2025.

[13] D. Wijendra, V. Jayasinghearachchi, O. A. P. Dilshan, H. M. K. C. B. Herath, Y. M. T. N. S. Yapa, and K. D. M. M. Rathnasiri, "OrchiZen: Hybrid integrated smart farming system for orchid plantations," in Proc. 2025 IEEE Int. Conf.

[14] Y. Sun, Y. Ming, X. Zhu, and Y. Li, "Out-of-distribution detection with deep nearest neighbors," in Proc. 39th Int. Conf. Machine Learning (ICML), Baltimore, MD, USA, 2022, pp. 20827-20840.

[15] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), Salt Lake City, UT, USA, 2018, pp. 4510-4520.

[16] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), Miami, FL, USA, 2009, pp. 248-255.

[17] N. A. C. Cressie, Statistics for Spatial Data, rev. ed. New York, NY, USA: Wiley, 1993.

[18] L. García, L. Parra, J. M. Jimenez, J. Lloret, and P. Lorenz, "IoT-based smart irrigation systems: An overview on the recent trends on sensors and IoT systems for irrigation in precision agriculture," Sensors, vol. 20, no. 4, p. 1042, 2020.

APPENDICES

The appendices carry only the material needed to verify the integrated solution. Component-level experimental records belong to the four individual theses.

Appendix A Integrated interface specification

This appendix records the request and response shapes of the interfaces listed in Table 4.1, so that the integration claims of Chapter 4 can be verified without reading the source.

The pollination assessment endpoint accepts a multipart request carrying a mandatory whole-plant image, an optional leaf close-up, and any traits the grower has explicitly corrected. Traits the grower has not answered are omitted from the request rather than sent with a placeholder value. The response carries the verdict, a confidence, and a trait-resolution block naming the origin of every value used. When the input gate refuses the image the endpoint returns HTTP 422 with a body containing an error code of not_an_orchid, a human-readable message and the gate's own measurements, which the application renders as a result rather than as a failure.

The compatibility endpoint accepts two parent names with the pod parent first, and optionally the Level 1 health assessment of either plant. It returns an evidence tier drawn from registered, genus proven, undemonstrated or blocked, the registered precedents supporting that tier, any warnings, and a statement of how each name was understood, whether its genus was written on the name, found in the knowledge base, or assumed. It does not return a probability of success under any circumstances.

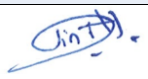
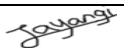
The logbook endpoints record what the grower attempted. A cross is created with both parent plants, each carrying a photograph and optionally a name; a plant with an unreadable tag is filed under an identifier instead, so that it can still be recorded. The cross is then advanced through the stages a capsule passes, or closed at the gate where it stopped. Records are written to local storage first and mirrored to the shared database second, so that an entry made in a shade house with no signal is not lost, and the counts endpoint returns numerators with their denominators rather than percentages until a gate has enough completed attempts behind it.

The section subtree owned by Component 3 is held under /tenants/{t}/, where {t} is the farm's tenant identifier. Under farm/houses/{h}, a house's meta record holds its name, dimensions, life cycle state (calibrating or active), master and pump channels, and each section under sections/{s} holds a meta record with its name, position, linked node and growth stage. Beside it, latest holds the most recent reading with temperature, humidity, light, VPD, a fault flag and a timestamp, plan holds today's watering time and duration with the inputs used, their source (measured or estimated) and any feed attached, and tray, fertilizer and control hold the tray decision and amount, the feed schedule, and the section's mode and pour lengths. Every reading is kept under farm/history/{h}/{s}, keyed by a time ordered identifier, and each master's queue, running pour, acknowledgements and stop request sit under farm/masters/{mac}. The accounts interface under /api/v2/accounts manages tenants, users and roles.

Appendix B Team contribution declaration

This appendix records the agreed division of work. The summary table is Table 4.2; this appendix carries the statement that all four members confirm it, together with any qualifications individual members wish to record.

We confirm that Table 4.2 is an accurate record of each member's contribution to the research, development, integration and evaluation of the integrated solution, and that the work presented in this common report is our own.

Team member	Student ID	Component	Signature	Date
Wickramasinghe D. P.	IT22065230	C1 - Hybrid pollination		10/10/2026
Rathnamalala D. M. J. P.	IT22064936	C2 - Disease detection		10/10/2026
Vithanage P. V. P. M.	IT22062642	C3 - Smart watering and fertilization		10/10/2026
Samadinee L. D. N.	IT22083500	C4 - Growth stage classification and bloom prediction		10/10/2026